Fiscal Year:	FY 2022	**Task Last Updated:**	FY 04/24/2024
PI Name:	Motil, Brian J. Ph.D.		
Project Title:	Analysis of Packed Bed Reactor-2 Experimental Data from International Space Station		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	44135	**Congressional District:**	10
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
Start Date:	06/25/2020	**End Date:**	12/24/2021
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date of CoI grant changed to 03/24/2023 per NSSC information (Ed., 11/2/22)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Balakotaiah, Vemuri Ph.D. (University of Houston grant 80NSSC20K0830)		
Grant/Contract No.:	Internal Project ; CoI grant 80NSSC20K0830		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of "Modeling and Analysis of Packed Bed Reactor Experimental Data from International Space Station" Principal Investigator Dr. Brian Motil ; CoInvestigator Vemuri Balakotaiah, Ph.D., grant 80NSSC18K0813 (formerly "Modeling and Experimental Studies on Gas-Liquid Two-Phase Flow Through Packed Beds in Microgravity" PI Dr. Brian Motil ; CoInvestigator Vemuri Balakotaiah, Ph.D., grant NNX14AD28G (and previously NNX10AL37G "Experimental Studies on Gas-Liquid Two-Phase Flow Through Packed Beds in Microgravity"), PI=Dr. Brian Motil; CoI=Dr. Vemuri Balakotaiah. Also included information for project “GAS-LIQUID TWO-PHASE FLOW THROUGH PACKED BEDS IN MICROGRAVITY,” internal project--Principal Investigator Dr. Brian Motil ; CoInvestigator Vemuri Balakotaiah, Ph.D., grant NNX17AF91G. A basic understanding of two-phase flows through porous media is of interest in many chemical and biological processing systems as well as many geophysical applications. NASA technologies include adiabatic two-phase flows or two-phase flows with heat transfer that are relevant to life support systems, fuel cells, in-situ resource utilization, heat pipes, materials processing, production of pharmaceutical grade water, and transport of nutrients in soils. A very important unit operation that involves gas-liquid flow through porous media is the fixed Packed Bed Reactor (PBR). Examples of current space systems in which a fixed PBR is used include the Volatile Removal Assembly (VRA), the Integrated Advanced Water Recovery System (AWRS), and IntraVenous Water GENeration system (IVGEN). NASA is currently operating systems with packed bed reactors for ground and flight-based testing with an incomplete understanding of how the reduced gravity environment affects the performance and reliability of these reactors. This is especially critical when gas-liquid flows are involved. The expected outcome of this research effort is to develop a set of guidelines and tools to enable engineers to reliably design and operate packed bed reactors for microgravity as well as the lunar and Martian environments. Existing approaches for predicting the flow regimes and pressure drop for gas-liquid flow through a PBR are valid only for 1 g, and are based on liquid (or gas) holdup within the bed. Initial microgravity tests conducted in the reduced gravity aircraft and drop tower clearly show that the liquid holdup is significantly influenced by gravity. Furthermore, most of these models have only been verified using the "trickle flow" regime which does not exist in microgravity. Basic hydrodynamic models developed specifically for reduced gravity have been partially validated within the constraints of aircraft and ground facilities. However, typical flow ranges used for real systems require up to several minutes to develop fully which is far beyond the time scales available for ground tests. In addition, instabilities and hysteresis effects that may exist require long testing durations to establish new operating conditions. Thus, testing on the International Space Station (ISS) is required for complete validation of the semi-empirical models. Flight experiments can provide critical hydrodynamic information for a project with a broader scope that includes reduced gravity aircraft and ground-based (1 g) experiments. The main objective of the overall project is to develop and validate macroscopic equations that can be used in partial and microgravity conditions to accurately predict flow pattern transitions; pressure drops; and mass transfer rates in chemical and biological reactors in which gas-liquid flows occur through randomly packed beds. The hydrodynamic investigations addressed in the Packed Bed Reactor Experiment (PBRE) focus on the transitions between flow regimes (e.g., bubbly-to-pulse flow transition) and the associated pressure gradients over the range of relevant operating regimes of the PBRs (e.g., liquid and gas flow rates, particle sizes, and fluid viscosities). These design tools provide a fundamental understanding of these phenomena in low gravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The PBRE provides a fundamental investigation of two phase flow through porous media in the absence of gravity. This enables a direct study of the other forces that impact the flow and reaction rates such as surface tension, inertia, and viscous forces. These new insights will impact reactor designs on Earth.
Task Progress:	From the Principal Investigators: • The pressure drop data from the International Space Station (ISS) Packed Bed Reactor Experiment (PBRE) microgravity experiments were analyzed in terms of a modified two phase friction factor which encompasses contribution from viscous, inertial, and capillary forces. • The capillary contribution to the pressure drop was found to be a strong function of the Capillary number. It was the dominant contribution, even at the highest gas and liquid flow rates studied. • Four flow regimes of large bubble, gas channeling, dispersed bubble, and pulse regimes were identified, and different friction factor correlations were developed for each regime. • The hysteresis effect was observed within the viscous capillary (V-C) regime under microgravity. A paper published in npj Microgravity provides a comprehensive summary of the PBRE findings: Taghavi, M., Motil, B.J., Nahra, H. et al. The international space station packed bed reactor experiment: capillary effects in gas-liquid two-phase flows. npj Microgravity 9, 55 (2023). https:// Note: Please see the Cumulative Bibliography for other articles and presentations derived from NASA support for this research (Ed., 4/24/2024).
Bibliography Type:	Description: (Last Updated: 04/26/2024)
Abstracts for Journals and Proceedings	Taghavi M, Motil B, Balakotaiah V. "Gas-liquid two-phase flow in permeable media in the absence of gravity." Gordon Research Conference: Flow and Transport in Permeable Media, Les Diablerets, Switzerland, July 17-22, 2022. Abstracts. Gordon Research Conference: Flow and Transport in Permeable Media, Les Diablerets, Switzerland, July 17-22, 2022. , Jul-2022
Abstracts for Journals and Proceedings	Nahra H, Taghavi M, Motil BJ, Balakotaiah V. "Gas-liquid two-phase flow in porous media under microgravity conditions of the International Space Station." International Space Station Research and Development Conference, Washington DC, July 2022. Abstracts. International Space Station Research and Development Conference, Washington DC, July 2022. , Jul-2022

Abstracts for Journals and Proceedings	Taghavi M, Balakotaiah V, Motil B, Nahra H. "Pressure gradient and gas hold-up in two-phase flows through porous media in microgravity." 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021. Abstracts. 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021. , Nov-2021
Articles in Peer-reviewed Journals	Taghavi M, Motil BJ, Nahra H, Balakotaiah V. "Gas-liquid flows through porous media in microgravity: Packed Bed Reactor Experiment-2." AIChE Journal. 2022 Aug;68(8):e17727. https://doi.org/10.1002/aic.17727 , Aug-2022
Articles in Peer-reviewed Journals	Taghavi M, Motil BJ, Nahra H, Balakotaiah V. "The international space station packed bed reactor experiment: capillary effects in gas-liquid two-phase flows." npj Microgravity. 2023 Jul 18;9(1):55. https://doi.org/10.1038/s41526-023-00302-2 , Jul-2023